

Detecting Purpose-Code Gaming in ISO 20022 Payments: A Synthetic Data-Quality Benchmark

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Abstract—ISO 20022 gives payment controls more structured data, but it also gives message creators choices about where names, addresses, and purpose information appear. This paper studies *purpose-code gaming*: deliberate changes intended to lower a modelled alert probability while preserving the payment’s business meaning. We define eleven manipulations across misfielding, code substitution, and message-path changes. A seeded generator creates synthetic pain.001 and pacs.008 records, adds ordinary data-quality defects, and labels adversarial edits. It enforces field-level constraints, serialises well-formed XML, and supports XSD validation when schemas are supplied. We also introduce screening-evasion lift (SEL), which measures how much a manipulation increases modelled pass-through. Across three 200,000-message runs, the best label-free detector reaches AUC 0.774 but catches only 9.9% of manipulated messages at an exact 1% alert capacity. A supervised model reaches 0.918 in-distribution, then falls from 0.815 to 0.366 on unseen misfielding. SEL ranges from 1.33 to 1.78. These results are synthetic and model-relative, but they show where message-level consistency checks help and where broader transaction context is needed.

Index Terms—ISO 20022, payment data quality, purpose codes, sanctions screening, anti-money laundering, adversarial evaluation, synthetic benchmark

I. INTRODUCTION

A payment message is both an instruction and a compliance record. Names, addresses, amounts, and purpose information may feed sanctions, fraud, and anti-money-laundering controls. The move to ISO 20022 makes much of that information easier to parse. In CBPR+, purely unstructured party and agent addresses cease to be supported from 14 November 2026; town and country must at least appear in designated fields [1]. CPMI’s harmonised data requirements similarly encourage consistent structured data and registered codes by end-2027, although CPMI is explicit that the requirements are guidance rather than a regulatory mandate [2].

Structure does not remove discretion. A message creator still chooses which optional fields to populate, which purpose code to declare, how much detail to place in free text, and sometimes which initiation or routing path is used. Screening is also selective. Published Swift guidance identifies elements that normally warrant direct screening and allows other fields to provide context instead [3]. Purpose and category-purpose codes may influence processing or monitoring, yet market-practice guidance notes that these codes are often misunderstood or

placed in the wrong field [4]. An ordinary data-quality error and a deliberate low-scrutiny choice can therefore leave similar evidence.

That overlap creates a practical evaluation problem. A detector can look good by finding malformed records, even though malformed records are mostly benign. It can also earn a high AUC by learning the edit signatures of one generator, then fail on a manipulation that was absent from training. The relevant question is narrower than “can the model spot unusual data?” It is whether deliberate changes remain distinguishable from routine error at an alert capacity a payment operation could actually use.

This paper makes three contributions. It defines eleven message-level manipulations and states the capability needed for each. It provides a seeded synthetic benchmark with a tunable rate of ordinary data-quality errors, five detector baselines, and a leave-one-family-out test. It also introduces screening-evasion lift (SEL), which separates the benefit gained by the adversary from the portion removed by a detector. The detectors are conventional on purpose; the benchmark, the operational measurement, and the failure cases are the main contribution. No real payment, customer, or institution-specific alert data is used.

II. BACKGROUND AND DESIGN GAP

A. Fields and control boundaries

The benchmark covers customer credit transfers at two stages: pain.001 at initiation and pacs.008 between financial institutions. Relevant content includes `Purp/Cd`, `CtgyPurp/Cd`, party names and postal addresses, `UltmtDbtr`, and structured or unstructured remittance information. Purpose describes the underlying reason for a transaction; category purpose is a higher-level instruction that may affect processing. Their exact use is corridor- and institution-specific, which is why the paper does not assign a universal risk meaning to any ISO code.

The address transition is useful context but not the attack itself. A hybrid address can retain free-text lines while placing town and country in their designated fields. That improves machine readability, but it does not guarantee that the most informative identity or location evidence sits in a field read by every control. Likewise, a valid purpose code can still be implausible for the creditor’s industry or inconsistent with the

remittance narrative. The benchmark treats these as consistency questions, not schema errors.

Selective screening is unavoidable. Reading every string as a party name would create an unmanageable false-positive burden, while ignoring contextual fields loses useful evidence. The tested controls therefore represent three layers: structural validation, consistency checks within one message, and comparison with a synthetic transaction history. This layering matters later: message-level checks can catch contradictions, but they cannot recover context that has been removed or shifted to a different path.

B. Related benchmarks and adversarial framing

Synthetic AML datasets usually operate at the transaction or graph level. AMLworld models realistic transaction networks with laundering labels, SAML-D supplies a labelled synthetic monitoring dataset, and strategic simulations allow laundering agents to adapt against a detector [5]–[7]. These resources study who pays whom, how much, and when. They do not encode the internal placement of ISO 20022 fields, so they cannot express a manipulation whose mechanism is that a name, code, or narrative appears in a different element. Transaction-purpose classification addresses enrichment of missing or noisy labels, rather than a label selected strategically [8].

Data-quality research also warns against attributing every performance change to the learning algorithm. Camacho and Rodríguez-Gómez show that dataset variants can move anomaly-detection results more than the model choice itself [9]. That finding motivates the explicit defect-rate parameter η in this benchmark. The optimisation follows standard cost-bounded adversarial classification and test-time evasion [10], [11]; the domain-specific part is the finite set of field and path changes that preserve the synthetic payment objective. We found no cited benchmark that combines those message-level operations with a controlled accidental-error floor. This is a scoped literature observation, not a claim of exhaustive priority.

III. THREAT MODEL AND MANIPULATIONS

The adversary can construct or amend a payment instruction but does not control a bank’s screening system, sanctions lists, or alert threshold. This covers a corporate originator, a payment-service provider (PSP), or an insider with legitimate initiation access. Message-path substitution is a stronger capability: it applies only to a PSP, bank-side actor, or originator that can choose among available channels. A typical corporate sender cannot simply select an interbank message type.

A black-box adversary knows public message guidance and receives only a coarse accept-or-hold outcome. A grey-box adversary also knows the relative ordering of the benchmark’s purpose-code weights. Neither version observes the exact threshold or a production rule set. The distinction is methodological: it lets the evaluation ask whether more knowledge makes the evader harder to catch, without suggesting that any particular bank exposes those weights.

Let x be an illicit synthetic payment instruction, m an admissible modification, $c(m)$ its operational cost, and $\pi(x)$

TABLE I
THE ELEVEN BENCHMARK PRIMITIVES. THE LABELS DESCRIBE THE SYNTHETIC OPERATION, NOT A CLAIM ABOUT EVERY BANK’S IMPLEMENTATION.

Family	Primitive operations
P1 misfielding	P1a move an ultimate-party name to remittance text; P1b destructure address content; P1c truncate a screened name; P1d omit the ultimate debtor.
P2 code choice	P2a replace purpose; P2b replace category purpose; P2c omit purpose; P2d create a code–narrative mismatch.
P3 structure	P3a use an alternate message path; P3b increase in-message splitting; P3c insert an intermediary representation.

the alert probability in the published screening abstraction. The evader chooses

$$m^* = \arg \min_m \{ \pi(m(x)) + \lambda c(m) \} \quad \text{s.t.} \quad V(m(x)), |m| \leq k, \quad (1)$$

where V requires the modified record to pass the benchmark’s structural and business checks and preserve the intended transfer. This is a cost-bounded, test-time evasion problem [10], [11]. It does *not* imply that every generated XML instance would be accepted or settled by a production bank.

P1 changes where identity or address evidence appears. P2 leaves the container intact but changes the declared semantics. P3 changes the message path or decomposition. These families describe the adversary’s action; they do not predict detectability. A manipulation is easier to detect when it leaves an internal contradiction, regardless of family.

IV. BENCHMARK AND METHOD

A. Synthetic population and edits

The generator builds debtors and creditors with an industry, home country, corridor, and pair-specific purpose profile. A benign message samples from that profile and populates party, address, purpose, and remittance fields. Pain.001 files are allowed to contain ordinary batches so that multiple transactions are not a perfect gaming indicator. In the default population, 30% of benign pain.001 files contain between 2 and 49 transactions.

A fraction η receives one ordinary data-quality defect. The default mix is 35% purpose content placed in remittance text, 30% partial address destructuring, 20% absent ultimate party, and 15% absent category purpose. These proportions are controlled benchmark choices, not measured industry rates. Their classes follow the types of field-use problems described by PMPG [4].

A fraction ρ is then marked illicit for the screening abstraction through a synthetic combination of party and corridor risk. For each such record, the evader enumerates beneficial primitive sets up to budget k . A perfectly rational minimum would concentrate on one cheap edit and leave too few samples for several primitives, so the implementation samples from a softmax over near-optimal responses. Ground truth records the chosen primitives, the alert probability before and after the edit, and any accidental defect already present.

TABLE II
CONSTRAINT GROUPS USED BY THE BENCHMARK.

ID	Check	Context
C1	Purpose-code membership	field
C2	Required structure and address tokens in free text	message
C3	Purpose-industry, batch, and ultimate-party plausibility	message
C4	Purpose code versus remittance narrative	message
C5	Pair-level purpose distribution versus prior history	history

B. Validation features and XML

Five constraint groups produce detector features. Table II separates local checks from those needing context. C1–C2 are intended to represent low-cost validation; C3–C4 look for contradictions within a message; C5 requires pair history. The purpose-history feature reflects the type of new-code and historical-pattern mismatch described by the Federal Reserve [12].

The field-level records can be serialised as pain.001.001.09 or pacs.008.001.08. The release checks XML well-formedness and exposes an optional XSD-validation hook when schemas are supplied. No XSD-validation outcome is used as experimental evidence, and the paper therefore does not claim that all emitted instances are production-valid. This distinction also avoids conflating an XSD pass with channel rules, business validation, or settlement.

C. Screening abstraction and calibration

The incumbent alert model is transparent rather than institution-specific. A name leg operates over a stated set of party fields. A score leg combines purpose, corridor, amount, and completeness, followed by a logistic threshold. A residual floor represents controls outside the evader’s reach. The combined probability is

$$\pi(x) = 1 - (1 - p_{\text{name}})(1 - p_{\text{score}})(1 - f). \quad (2)$$

The weights are illustrative and published with the code. They are not estimates of any bank’s rule set.

Calibration deserves explicit treatment because P1 mainly attacks the name leg and P2 mainly attacks the score leg. Setting the desired baseline alert probability π_0 by adjusting only one leg can make the other family irrelevant. The benchmark instead divides baseline survival multiplicatively between the two legs using a stated name-share parameter σ . The main setting uses $\pi_0 = 0.45$ and $\sigma = 0.6$; a 3×3 grid varies $\pi_0 \in \{0.30, 0.45, 0.60\}$ and $\sigma \in \{0.4, 0.6, 0.8\}$.

D. Detectors and evasion metrics

D_0 applies C1–C2 checks. D_1 adds fixed C3–C4 rules. D_2 is a supervised gradient-boosted model over one-hot codes, corridor, amount, and completeness features. D_3 is an isolation forest fitted to benign C1–C5 features [13]. D_4 sums standardised D_1 and D_3 scores. Only D_2 trains on gaming labels.

TABLE III
DETECTION RESULTS. PANEL (A) IS THE MEAN OF THREE 200,000-RECORD RUNS. TPR IS EVALUATED AT AN EXACT ALERT CAPACITY WITH LABEL-BLIND TIE-BREAKING. PANEL (B) USES PURE-FAMILY RECORDS AT $k = 1$; ONLY D_2 IS RETRAINED.

(a) Main setting				
Detector	Labels	AUC	TPR@1%	TPR@0.1%
D_0 C1–C2 checks	no	0.497	0.009	0.001
D_1 fixed rules	no	0.750	0.105	0.020
D_2 supervised	yes	0.918	0.201	0.020
D_3 anomaly	no	0.752	0.040	0.007
D_4 combined	no	0.774	0.099	0.018

(b) Leave-one-family-out				
Family	n	D_1	D_2 : in $\rightarrow$ out	D_4
P1 misfiling	3980	0.733	0.815 $\rightarrow$ 0.366	0.744
P2 code choice	1728	0.843	0.989 $\rightarrow$ 0.779	0.800
P3 structure	3917	0.445	0.689 $\rightarrow$ 0.487	0.484

AUC measures ranking but not the value of an attack. Let π_0 be the alert probability before manipulation and π_m after it. We define

$$\text{SEL}(m) = \frac{1 - \pi_m}{1 - \pi_0}. \quad (3)$$

If detector D alerts manipulated and unmanipulated illicit payments at rates δ_m and δ_0 , the residual lift is

$$\text{SEL}_{\text{res}}(m, D) = \frac{(1 - \pi_m)(1 - \delta_m)}{(1 - \pi_0)(1 - \delta_0)}. \quad (4)$$

SEL of 1 means that the edit provides no modelled benefit. The gap between SEL and residual SEL is the portion removed by the detector.

E. Evaluation protocol

The main setting uses 200,000 records per seed, seeds 7–9, $\eta = 0.15$, $\rho = 0.05$, $k = 2$, $\lambda = 0.18$, softmax temperature 0.10, residual floor 0.10, target $\pi_0 = 0.45$, and a 0.6 name-leg share. We report AUC and true positive rate at alert capacities of 1% and 0.1%.

Discrete detector scores create ties. A plain “score above quantile” rule can therefore alert far more records than the stated capacity. The implementation selects every score above the boundary and assigns the tied block the label-blind fraction needed to fill the capacity exactly. At 1% of 200,000 records this is 2,000 expected alerts, not all records tied with the 2,000th. Reported TPR is the expectation under uniform tie-breaking.

For generalisation, D_2 is retrained while one primitive family is withheld. The evaluation uses $k = 1$ and pure-family messages, preventing a held-out family from leaking through a second primitive. Its held-out AUC is compared with an in-distribution AUC on the same rows. The protocol also varies η , k , adversary knowledge, and the calibration of π . Configuration, seeds, and raw JSON outputs are retained with the release.

V. RESULTS

Table III(a) shows a large gap between ranking and useful recovery. D_4 separates manipulated from ordinary records with AUC 0.774, yet it finds only 9.9% of manipulations when

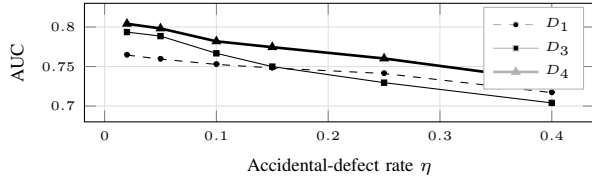


Fig. 1. AUC as the ordinary defect rate rises. Each point uses 100,000 records at seed 7; only AUC, which is unaffected by threshold ties, is shown.

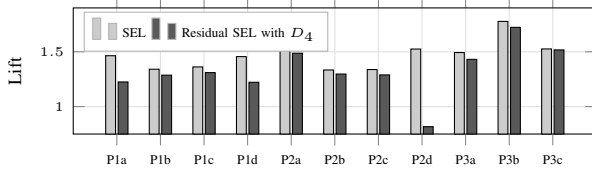


Fig. 2. Mean screening-evasion lift before and after D_4 across three seeds. Values are relative to the published screening abstraction.

exactly 1% of all records may be alerted. At 0.1% capacity, recovery is 1.8%. The supervised D_2 reaches AUC 0.918 and doubles the 1% recovery, but it assumes labels that are unlikely to exist for new manipulations.

Table III(b) shows how brittle that supervised advantage can be. When P1 is absent from training, D_2 falls from 0.815 to 0.366 on the same evaluation rows. Its P2 drop is smaller but still material. P3 is not usefully separated by the tested message-level controls: D_4 scores 0.484 in the pure-family test and D_1 scores 0.445. These numbers do not establish that P3 is inherently hidden. They show that the current fields and features provide little direction for ranking it.

The noise experiment in Fig. 1 shows gradual degradation rather than collapse. Raising η from 0.02 to 0.40 lowers D_4 AUC by 0.072. The fixed rules are the most stable. The anomaly model degrades faster because the benign distribution it learns is itself moving. At the highest noise rate, D_1 and D_4 still rank above chance, but the main operational result warns against treating that ranking as sufficient.

Manipulation increases modelled pass-through by 33%–78% in the main calibration. The detector’s value varies sharply by primitive. For the code–narrative mismatch P2d, residual lift falls below 1 ($1.525 \rightarrow 0.817$), so the added check more than removes the simulated benefit. For in-message splitting P3b, it barely changes ($1.777 \rightarrow 1.723$). P3c is even more stark in detection rate: only about 0.6% of its selected instances are alerted by D_4 at the 1% capacity. Residual SEL should nevertheless be read cautiously because the unevaded illicit baseline contains only 57–60 records per seed in this configuration.

Across the 3×3 calibration grid, SEL ranges from 1.13 to 2.59 while D_4 AUC remains between 0.772 and 0.795. The size of the modelled threat is therefore sensitive to the assumed baseline and name-share, while the relative detection difficulty is more stable. This is why the release reports the grid rather than treating one SEL number as a deployment estimate.

Table IV supports the same interpretation. More edits make

TABLE IV
 D_4 ABLATIONS, 100,000 RECORDS AT SEED 7 PER SETTING.

Factor	Setting	AUC
Primitive family	P1 / P2 / P3	0.794 / 0.834 / 0.548
Evader budget	$k = 1$ / $k = 2$ / $k = 3$	0.647 / 0.774 / 0.844
Knowledge	black / grey	0.738 / 0.774

the evader easier to rank because each edit can leave another inconsistency. The grey-box adversary is also easier to detect than the black-box one: acting on code-weight knowledge touches fields that C3–C4 examine. A more informed adversary is therefore not automatically a stealthier one in this benchmark.

VI. DISCUSSION AND LIMITATIONS

The practical lesson is not that structured data weakens controls. It gives controls better inputs and enables useful consistency checks. The narrower lesson is that validation, screening coverage, and behavioural monitoring must be designed together. C1–C2 catch malformed or incomplete data. C3–C4 catch contradictions within one message. C5 or other longitudinal features are needed when the message is internally plausible. Changes to path, batch structure, or party chain may require channel history, agent-chain context, or comparison with the originator’s normal files. A message-only model cannot infer evidence that the message does not contain.

The study also separates detector credit from incumbent-control credit. A primitive can be easy to classify yet offer little extra protection if the incumbent model would already alert it. Conversely, modest AUC can still be useful for a narrow primitive if it removes substantial lift at the operating threshold. SEL and residual SEL make that distinction visible, but they inherit the assumptions of π .

Several limitations remain. The screening abstraction is transparent and reproducible, not a reconstruction of a bank’s engine, so SEL is a stress measure rather than an estimate of losses or real evasion. The distributions of purpose codes, defects, batching, and illicit traffic are assumptions. The generator and detectors also share a feature vocabulary, which can reward recognition of implementation artefacts; the leave-one-family-out collapse of D_2 demonstrates the risk rather than solving it. External validity requires recalibration on an institution’s own traffic.

The XML layer has a similarly narrow claim. The serializers are checked for well-formedness, the pain.001 ultimate-debtor ordering was corrected during review, and the harness accepts external XSDs. The release does not treat XSD validity as proof of business acceptance. P3a is also capability-dependent and should not be attributed to an ordinary corporate originator without route choice.

The corpus uses synthetic entities and illustrative weights. It contains no customer records, employer data, production alert outcomes, or corridor-tuned control parameters. The primitive taxonomy is published for defensive coverage analysis: an institution can map the families to its own screened fields, message paths, and history features without disclosing payment

data. A stronger next experiment would calibrate π against anonymised alert outcomes inside one institution and evaluate path-aware controls without exporting raw messages.

VII. CONCLUSION

Purpose-code gaming remains distinguishable from routine data-quality error in this benchmark, but good AUC does not translate into strong recovery at a small alert capacity. Message-level checks work best when an edit creates a contradiction, as P2d does, and poorly when the edit changes the container while remaining internally coherent. The supervised model’s failure on an unseen family reinforces the same point: recognition of known edit signatures is not the same as coverage of the underlying risk. The release provides a compact way to test that boundary, together with the assumptions needed to challenge it.

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